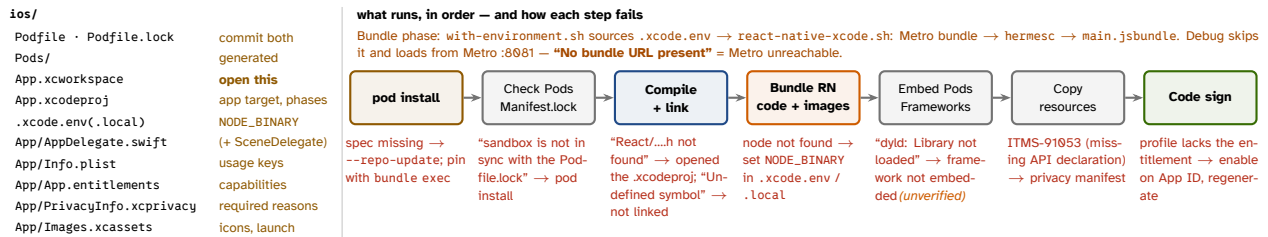


**In one line:** The iOS app is an ordinary **Xcode project** whose dependencies come from **CocoaPods** (pod install turns node\_modules podspecs into Pods.xcodeproj), whose **build phases** bundle your JS, and whose **Info.plist, entitlements, privacy manifest and signing** are yours to get right — most “RN build errors” are one of those phases failing.



## Podfile and pod install

```

platform :ios, min_ios_version_supported # 15.1 in 0.85
prepare_react_native_project!
linkage = ENV['USE_FRAMEWORKS'] # static / dynamic
use_frameworks! :linkage => :linkage.to_sym if linkage
target 'App' do
  config = use_native_modules! # autolinking
  use_react_native!(:path => config[:reactNativePath],
    :app_path => "#{Pod::Config.instance.installation_root}/..")
  post_install do |installer|
    react_native_post_install(installer, config[:reactNativePath])
  end
end
end

```

- pod install resolves every native dep's podspec, runs **Codegen**, writes Pods.xcodeproj + the **workspace**. Any native dep change, RN upgrade or Podfile edit ⇒ cd ios && bundle exec pod install (the Gemfile pins CocoaPods).
- use\_frameworks! (CocoaPods default linkage: dynamic) builds pods as frameworks; RN Firebase on CocoaPods needs :linkage => :static (it now defaults to SwiftPM). Unset, pods are static libraries.
- Since **0.84** the RN core is **precompiled** (RCT\_USE\_PREBUILT\_RNCORE=0 builds from source); you can't step into RN internals.
- CocoaPods trunk goes **read-only on 2 Dec 2026**; RN **0.87** adds experimental, opt-in **SwiftPM** (npx react-native spm) — CocoaPods stays default.

## AppDelegate — Swift since 0.77

One AppDelegate.swift replaced main.m + AppDelegate.h/.mm. It builds a RCTReactNativeFactory(delegate:) whose ReactNativeDelegate (RCTDefaultReactNativeFactoryDelegate) returns bundleURL() — Metro in DEBUG, main.jsbundle otherwise — sets dependencyProvider = RCTAppDependencyProvider() (0.77+: forget it and third-party modules misbehave), then startReactNative(withModuleName:in:). C++ TurboModules still need an ObjC++ AppDelegate. **iOS 27 SDK requires the UIWindow life cycle** (WWDC26, TN3187: built with Xcode 27 without it, the app won't launch); the template adopted a SceneDelegate in Sep 2026 — deep links move to scene(:openURLContexts:) → RCTLinkingManager.

## Privacy manifest

PrivacyInfo.xcprivacy: required-reason APIs (template: FileTimestamp C617.1, UserDefaults CA92.1, SystemBootTime 35F9.1), collected data types, tracking domains. SDKs on Apple's list ship their own **signed** manifest; Xcode's privacy report aggregates them. Since 1 May 2024 App Store Connect refuses uploads missing a reason (ITMS-91053).

## Info.plist — the keys you will be asked about

- Usage descriptions:** NSCameraUsageDescription, NSMicrophoneUsage..., NSPhotoLibraryUsage... (+ ...AddUsage...), NSLocationWhenInUseUsage..., NSLocationAlwaysAndWhenInUseUsage..., NSFaceIDUsage..., NSBluetoothAlwaysUsage..., NSContactsUsage..., NSUserTrackingUsage... (ATT). Missing → “the system **terminates your app**” (Apple) at first access; vague text → review rejection (unverified). A JS permission library does not add them — you (or a config plugin) do.
- Also: CFBundleURLTypes (schemes), LSApplicationQueriesSchemes (canOpenURL list), UIBackgroundModes, UIAppFonts, ITSAppliesNonExemptEncryption, ATS.

## Capabilities = entitlement + App ID + profile

A capability (Signing & Capabilities tab) is three things that must agree: a key in App.entitlements · the feature enabled on the **App ID** · a **provisioning profile** that contains it. Keys: aps-environment (push), com.apple.developer.associated-domains (applinks:example.com), com.apple.security.application-groups, keychain-access-groups. **Signing:** automatic (team id, Xcode manages profiles) or manual (fastlane match in CI); dev vs distribution cert; bundle id = App ID.

## Reset recipe (in this order)

Clean Build Folder (Shift-Cmd-K) → delete ios/build + DerivedData → rm -rf ios/Pods → bundle exec pod install → npx react-native start --reset-cache. Script phase fails with “Sandbox: rsync deny” → ENABLE\_USER\_SCRIPT\_SANDBOXING = NO (unverified). Xcode 27 fails pods whose deployment target is < 15.0 — raise it in post\_install (unverified).

## Traps

- ! Opening App.xcodeproj: no Pods → “file not found”.
- ! Metro reload after adding a native lib: needs pods + rebuild.
- ! Hand-editing ios/ in Expo CNG: prebuild overwrites it.
- ! Permission asked from JS, no Info.plist key → crash.

## Remember

**Workspace, not project · pod install after native change · plist, entitlements, manifest, signing.**